

Executive Project Report: Power BI Sales & Customer Performance Analytics

Project Title: Sales & Customer Performance Dashboard

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Deliverable Type: End-to-End Business Intelligence & Analytical Report

1. Executive Summary

This report documents the architectural design, business logic, visualization structure, and data model for the **Sales & Customer Performance Dashboard** created in Power BI. The primary goal of this analytics solution is to centralize high-level financial metrics, transactional volumes, customer engagement indicators, and product return analytics into an interactive reporting framework.

Through a star-schema data architecture and Data Analysis Expressions (DAX) modeling, the dashboard allows business stakeholders to monitor real-time sales metrics, analyze multi-dimensional sales distributions, evaluate regional market penetration, and identify operational bottlenecks within return workflows.

2. Dataset & Data Model Architecture

The data architecture is built upon six foundational data entities designed to track customer interactions, order processing, product hierarchies, pricing structures, and geographic segmentation.

2.1 Core Data Schema

Entity / Data Area	Description	Primary Analytical Function
Customers	Demographic & customer activity records	Key source for customer acquisition, distinct counts, and active user tracking.
Orders	Transaction-level order logs	Captures total order counts, order frequency, and transactional line items.
Products	Categorical hierarchy and item attributes	Enables product, category, and subcategory dimension slicing.
Sales	Monetization, unit volume, and pricing data	Basis for core financial metrics (Total Revenue, Average Selling Price).
Returns	Post-purchase log of returned items and feedback	Tracks defect rates, return volume, and dissatisfaction drivers.
Regions	Geographic segmentation tables	Facilitates regional comparative analysis and territory performance evaluation.

3. Key Performance Indicators (KPI Overview)

The central metrics generated by the dashboard across all evaluated business periods are structured as follows:

- **Total Customers:** 200 registered individual customer accounts.
- **Active Customers:** 198 actively purchasing customers within the primary evaluation period.
- **Total Order Volume:** ~1,000 distinct orders processed.
- **Total Gross Revenue:** \$844.02K accumulated across all sales channels.
- **Average Selling Price (ASP):** \$152.93 per unit sold.
- **Total Product Returns:** 50 product return cases logged across categories.

4. Multi-Page Dashboard Analysis & Visual Layout

The report interface consists of dedicated analytical pages designed to serve executive leadership, inventory/sales managers, and customer success teams.

Page 1: Executive Overview & Geographic Sales Distribution

Top KPI Cards

- **Total Customers:** 200
- **Active Customers:** 198
- **Total Orders:** 1K
- **Total Sale:** \$844.02K
- **Average Price Per Unit:** \$152.93
- **Total Return:** 50

Visualizations & Functional Breakdown

1. Total Sale by Category (Pie Chart)

- **Distribution:** Office Supplies accounts for **35.92%** (\$303.15K), Furniture accounts for **32.53%** (\$274.57K), and Technology accounts for **31.55%** (\$266.30K).
- **Business Impact:** Revenue is well-balanced across major categories, with Office Supplies generating the highest individual share.

2. Total Customers by Region (Donut Chart)

- **Distribution:** South: **54 (27%)**, West: **51 (25.5%)**, North: **50 (25%)**, East: **45 (22.5%)**.
- **Business Impact:** Demonstrates balanced customer acquisition, with the South region leading overall volume.

3. Total Sale by Region (Clustered Bar Chart)

- **Distribution:** Regional ranking mirrors customer distribution, with **South** generating the largest sales volume, followed by **North**, **West**, and **East**.

- **Business Impact:** Provides clear regional revenue baselines for territory resource allocation.
4. **Total Return by Category (Column Chart)**
- **Distribution:** **Office Supplies** records the highest return rate (~20 items), followed by **Furniture** (~15 items) and **Technology** (~12 items).
 - **Business Impact:** Highlights a potential product quality or listing discrepancy within the Office Supplies vertical.
5. **Total Sale Trend by Year and Month (Line Chart)**
- **Trend Analysis:** Tracks performance from **July 2024 through May 2025**, highlighting seasonal peaks in November 2024 and March 2025 (~\$60K+) followed by dips in August 2024 and February 2025.
 - **Business Impact:** Assists in inventory planning and revenue forecasting based on historical demand cycles.

Page 2: Product & Transactional Level Analytics

Top KPI Cards

- **Total Sale:** \$53.30K (Filtered view)
- **Products Sold:** 52 units
- **Average Price Per Unit:** \$149.24
- **High Volume Revenue:** \$36.60K
- **High Units Sold:** 29.62K

Visualizations & Functional Breakdown

1. **Total Sale by Day (Line Chart)**
 - Captures day-by-day sales variance within a 30-day window, indicating major spikes around **Day 18** and **Day 28** (peaking above \$6K).
2. **Total Sale by SubCategory (Donut Chart)**
 - Breakdown of top sub-categories: **Chairs** (20.24% / \$10.79K), **Phones** (18.82% / \$10.03K), **Paper** (16.36% / \$8.72K), and **Binders** (15.41% / \$8.21K).
3. **Top 10 High Sale Products (Data Table)**
 - Ranks individual product SKUs by revenue contribution, led by *Revolutionize Cross-Media Channels* (\$3,938.29), *Monetize Proactive Niches* (\$2,752.08), and *Redefine Robust ROI* (\$2,227.36). Top 10 items account for **\$20,874.62** in combined sales.

Page 3: Customer Engagement & Return Drivers

Visualizations & Functional Breakdown

1. **Active Customers by Day (Line Chart)**

- Tracks daily customer active counts, consistently fluctuating between 10 and 40 active users per day.

2. Active Customers by Category & Region (Donut Charts)

- **Category Active Share:** Furniture (**33.81%**), Office Supplies (**33.40%**), Technology (**32.79%**).
- **Region Active Share:** South (**27.27%**), North (**25.25%**), West (**24.75%**), East (**22.73%**).

3. Top 10 High Value Customers (Data Table)

- Details core account performance, highlighting top revenue contributors:
 - **Brent Green:** \$11,445.62
 - **Michael Rich:** \$11,185.28
 - **Thomas Taylor:** \$10,877.71
 - **Total Revenue (Top 10):** \$99,617.01

5. DAX Measures Documentation

The business logic and dynamic calculation model were built using the following DAX measures:

Measure Name	Analytical Function & Business Logic
Active customers	Calculates distinct count of active customers within the active date range.
All Product Sales	Computes overall sales across all categories, overriding local visual context.
Average Price Per Unit	Evaluates the mean unit pricing across transactions (Total Revenue / Volume).
High Volume Revenue	Filters and sums total sales generated strictly from high-volume transaction orders.
High_units_sold	Aggregates quantities from transactions exceeding bulk volume limits.
Premium_transtion_Count(>1000)	Counts distinct transactions with individual order values exceeding \$1,000.
Product_Sold	Computes total quantity of product units sold.

Measure Name	Analytical Function & Business Logic
sales last year	Time-intelligence measure retrieving equivalent period sales from the prior year.
Total Orders	Counts the total number of orders placed.
Total_Customers	Measures unique customer ID entries across the enterprise dataset.
Total_Furniture_Sale	Calculates revenue specifically generated by the Furniture product category.
Total_Product	Tracks total unique active product SKUs in the portfolio.
Total_Return	Aggregates total instances of returned items.
Total_Sale	Evaluates absolute gross sales revenue.
YTD	Accumulates Year-To-Date revenue totals from the start of the calendar year.

6. Strategic Business Insights & Recommendations

1. **Address Return Rates in Office Supplies:** Office Supplies yields the highest individual revenue share (\$303.15K), but also accounts for **40% of total returns**. Conducting supplier quality audits on high-return items will reduce operational fulfillment costs.
2. **Targeted Expansion in East Territory:** The East region represents the lowest sales volume (\$0.2M) and customer count (22.5%). Increasing targeted promotional activity in the East can help balance territorial market share.
3. **Key Account Loyalty Program:** The top 10 accounts contribute **\$99,617.01** (~11.8% of total enterprise revenue). Implementing a VIP customer tier for accounts generating >\$10,000 will strengthen customer retention and prevent churn among high-value buyers